Recapture Analsyis: COLONY

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Recapture Analysis in COLONY

The COLONY job ran successfully in ~10 hours (using the parallel COLONY executable with 8 cores). Note that this analysis is faster when using the monogamous mating model, which Wang (2016) recommends. This way, only full-sib groups are identified at first. It is unlikely that pairs of samples identified as half-sibs would actually be recaptures. While this is fine for the clone analysis in COLONY, it would not be appropriate for an analysis intended to reconstruct the pedigree of the sample or estimate N_b using the sibship method implemented in COLONY. Those analyses would require the polygamous mating model.

In this markdown, we read in the outputs from COLONY and calculate a few basic summaries...

```
col.out <- read.table("/home/jdrobins/CGA_BlackBears/recaps/CGA.BestClone",
                      header = TRUE)

col.list <- vector("list", nrow(col.out))
for(i in 1:nrow(col.out)) {
  col.list[[i]] <- strsplit(col.out[i,3], split=",")[[1]]
}

col.list[[1]]
```

```
## [1] "UGA_173101_P020_WD02" "UGA_173101_P018_WE05" "UGA_173101_P018_WA11"
## [4] "UGA_173101_P017_WC11" "UGA_173101_P020_WC09" "UGA_173101_P021_WD11"
```

```
col.list[[2]]
```

```
## [1] "UGA_173101_P020_WC12"
```

```
col.list[[3]]
```

```
##  [1] "UGA_173101_P020_WC08" "UGA_173101_P019_WB08" "UGA_173101_P019_WD06"
##  [4] "UGA_173101_P016_WF02" "UGA_173101_P016_WD09" "UGA_173101_P016_WE06"
##  [7] "UGA_173101_P015_WF08" "UGA_173101_P017_WA03" "UGA_173101_P023_WC06"
## [10] "UGA_173101_P023_WG02" "UGA_173101_P021_WB10" "UGA_173101_P020_WF12"
## [13] "UGA_173101_P021_WC12" "UGA_173101_P021_WD04" "UGA_173101_P022_WA01"
```

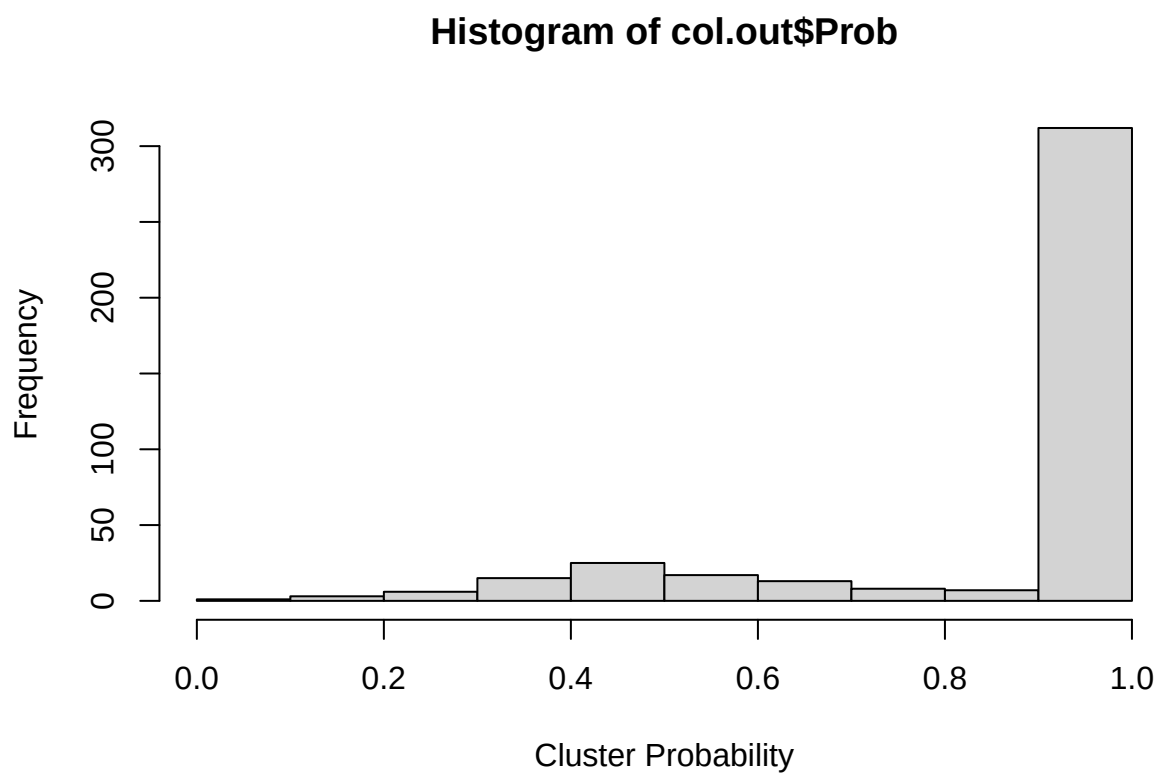
Summarize output

Some quick summaries of the clusters identified by `COLONY`...

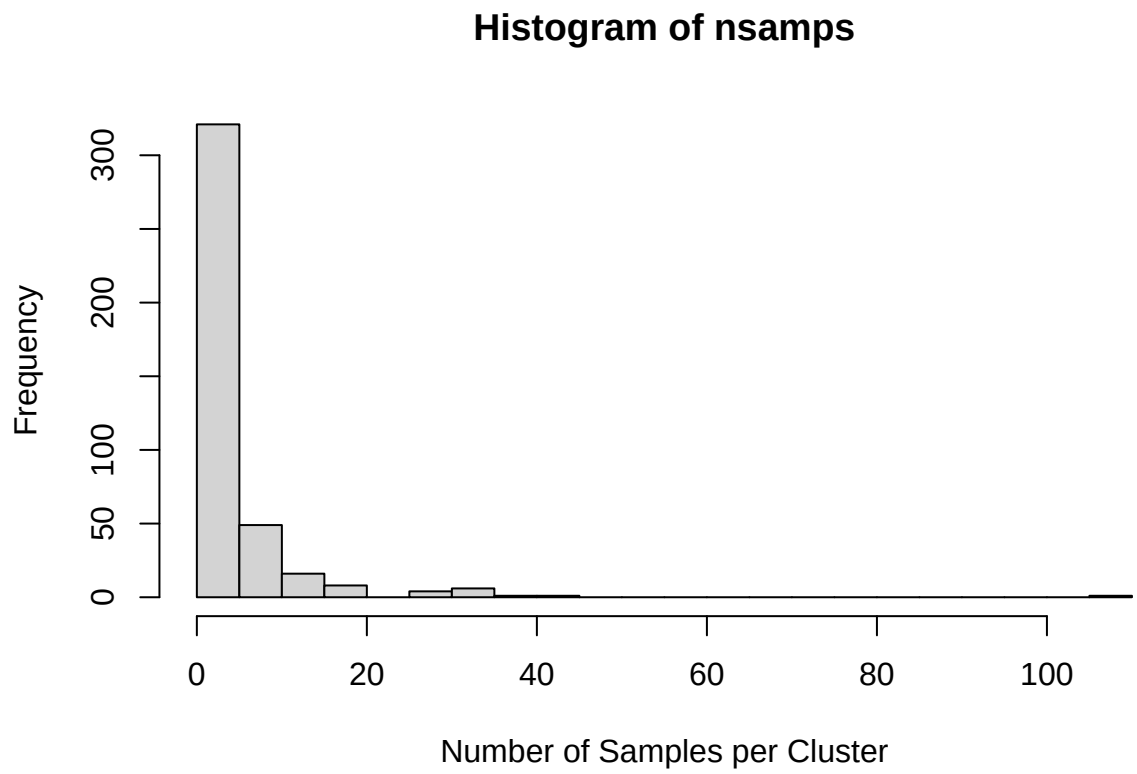
```
#Total number of unique genotypes
length(col.list)
```

```
## [1] 407
```

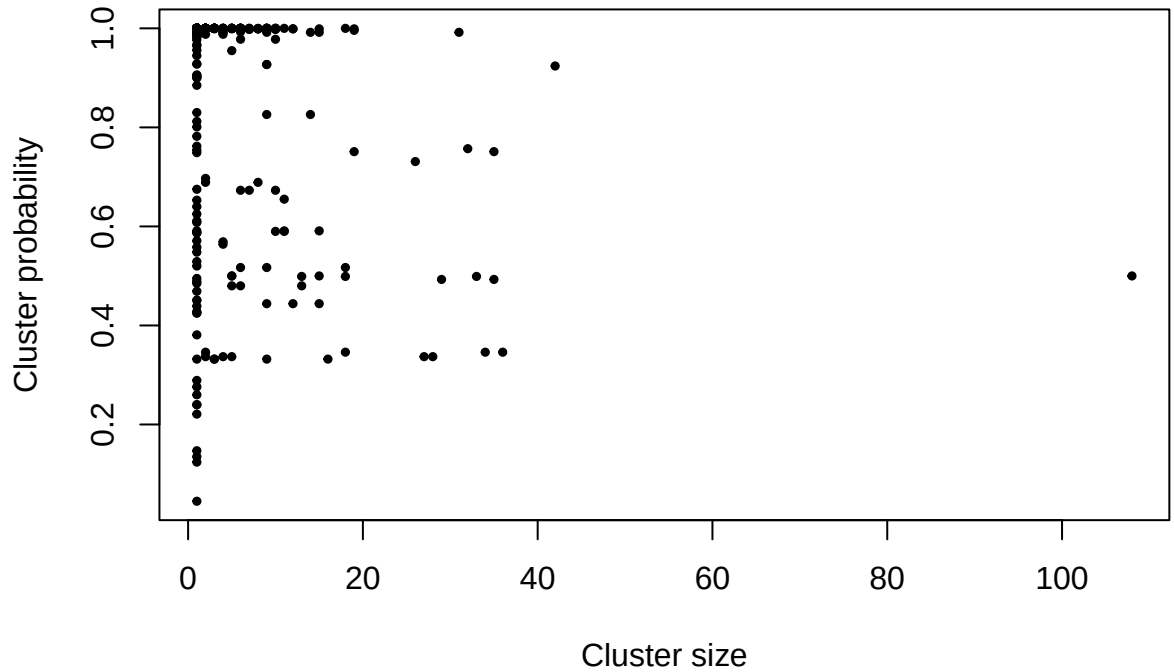
```
#Histogram of cluster probabilities
hist(col.out$Prob, xlab = "Cluster Probability")
```




```
#Sizes of clusters
nsamps <- sapply(col.list, FUN=length)
hist(nsamps, xlab = "Number of Samples per Cluster", breaks = 20)
```




```r
#Relationship between cluster size and probability?
plot(x = nsamps, y=col.out$Prob,
     xlab = "Cluster size", ylab = "Cluster probability",
     pch = 19, cex = 0.5)
```




```r
cor.test(nsamps, col.out$Prob)
```

```
##
##  Pearson's product-moment correlation
##
## data:  nsamps and col.out$Prob
## t = -5.7499, df = 405, p-value = 1.761e-08
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.3622523 -0.1823885
## sample estimates:
##        cor
## -0.2747219
```

Further analysis, including assignment of cloneIDs and comparisons of outputs from `COLONY` and `PLINK2` are in `10_recapFinal`.